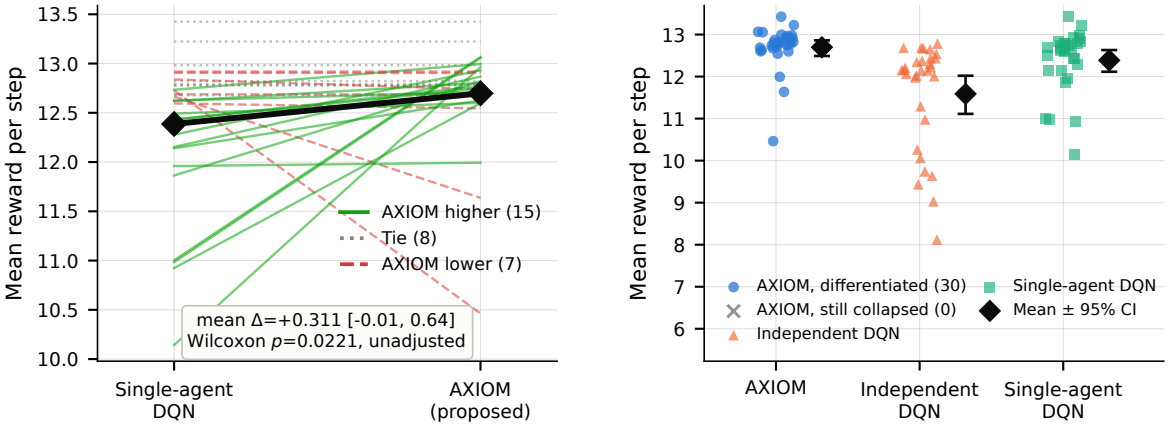

(a) Paired per-seed comparison (n=30)
Mean reward per step
13.5
13.0
12.5
12.0
11.5
11.0
10.5
10.0
AXIOM higher (15)
Tie (8)
AXIOM lower (7)
mean Δ=+0.311 [-0.01, 0.64]
Wilcoxon p=0.0221, unadjusted
Single-agent DQN
AXIOM (proposed)
(b) Per-seed distribution (n=30/arm)
Mean reward per step
13
12
11
10
9
8
7
6
AXIOM, differentiated (30)
AXIOM, still collapsed (0)
Independent DQN
Single-agent DQN
Mean ± 95% CI
AXIOM
Independent DQN
Single-agent DQN